

武器系统动力学建模与仿真

刚体的空间运动 -I

发射动力学研究所

南京理工大学

2025

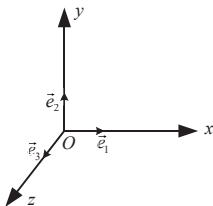
刚体的空间运动 -I

- ① 坐标系、坐标轴上的单位矢量与矢量基
- ② 矢量及矢量运算的坐标表示
- ③ 不同坐标系矢量基之间的关系与方向余弦矩阵
- ④ 刚体的定点简单转动
- ⑤ 定点简单转动与方向余弦矩阵的关系
- ⑥ 欧拉参数

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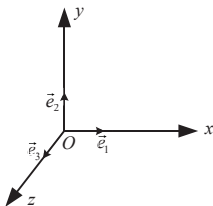
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右手笛卡尔直角坐标系及其坐标轴上的单位矢量



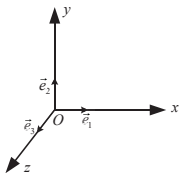
$$\left\{ \begin{array}{lll} \vec{e}_1 \cdot \vec{e}_1 = 1; & \vec{e}_1 \cdot \vec{e}_2 = 0; & \vec{e}_1 \cdot \vec{e}_3 = 0; \\ \vec{e}_2 \cdot \vec{e}_1 = 0; & \vec{e}_2 \cdot \vec{e}_2 = 1; & \vec{e}_2 \cdot \vec{e}_3 = 0; \\ \vec{e}_3 \cdot \vec{e}_1 = 0; & \vec{e}_3 \cdot \vec{e}_2 = 0; & \vec{e}_3 \cdot \vec{e}_3 = 1. \end{array} \right.$$

右手笛卡尔直角坐标系及其坐标轴上的单位矢量



$$\left\{ \begin{array}{lll} \vec{e}_1 \times \vec{e}_1 = 0; & \vec{e}_1 \times \vec{e}_2 = \vec{e}_3; & \vec{e}_1 \times \vec{e}_3 = -\vec{e}_2; \\ \vec{e}_2 \times \vec{e}_1 = -\vec{e}_3; & \vec{e}_2 \times \vec{e}_2 = 0; & \vec{e}_2 \times \vec{e}_3 = \vec{e}_1; \\ \vec{e}_3 \times \vec{e}_1 = \vec{e}_2; & \vec{e}_3 \times \vec{e}_2 = -\vec{e}_1; & \vec{e}_3 \times \vec{e}_3 = 0. \end{array} \right.$$

右手笛卡尔直角坐标系及其坐标轴上的单位矢量



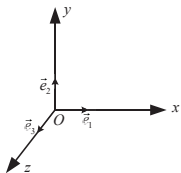
$$\vec{e}_m \cdot \vec{e}_n = \delta_{mn}$$

$$\vec{e}_m \times \vec{e}_n = \sum_{l=1}^3 \varepsilon_{mnl} \vec{e}_l =: \varepsilon_{mnl} \vec{e}_l$$

$$\delta_{mn} = \begin{cases} 1 & m = n \\ 0 & m \neq n \end{cases}$$

$$\varepsilon_{mnl} = \begin{cases} 1 & (m, n, l) = (1, 2, 3); (2, 3, 1); (3, 1, 2) \\ -1 & (m, n, l) = (2, 1, 3); (3, 2, 1); (1, 3, 2) \\ 0 & m = n; \quad n = l; \quad l = m \end{cases}$$

右手笛卡尔直角坐标系矢量基及其运算法则



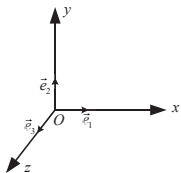
右手笛卡尔直角坐标系矢量基

$$\vec{e} \triangleq \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}, \quad \vec{e}^T = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix}.$$

坐标系矢量基运算法则

$$\begin{aligned} \vec{e}^T \cdot \vec{e} &= \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix} \cdot \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix} \\ &= \vec{e}_1 \cdot \vec{e}_1 + \vec{e}_2 \cdot \vec{e}_2 + \vec{e}_3 \cdot \vec{e}_3 \\ &= 3 \end{aligned}$$

右手笛卡尔直角坐标系矢量基及其运算法则



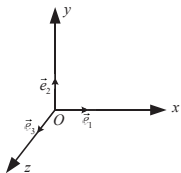
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坐标系矢量基运算法则

$$\vec{e} \cdot \vec{e}^T = \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix} \cdot \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix} = \begin{bmatrix} \vec{e}_1 \cdot \vec{e}_1 & \vec{e}_1 \cdot \vec{e}_2 & \vec{e}_1 \cdot \vec{e}_3 \\ \vec{e}_2 \cdot \vec{e}_1 & \vec{e}_2 \cdot \vec{e}_2 & \vec{e}_2 \cdot \vec{e}_3 \\ \vec{e}_3 \cdot \vec{e}_1 & \vec{e}_3 \cdot \vec{e}_2 & \vec{e}_3 \cdot \vec{e}_3 \end{bmatrix} = \mathbf{I}_3$$

右手笛卡尔直角坐标系矢量基及其运算法则



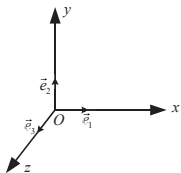
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坐标系矢量基运算法则

$$\begin{aligned} \vec{e}^T \times \vec{e} &= \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix} \times \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix} \\ &= \vec{e}_1 \times \vec{e}_1 + \vec{e}_2 \times \vec{e}_2 + \vec{e}_3 \times \vec{e}_3 \\ &= \vec{0} \end{aligned}$$

右手笛卡尔直角坐标系矢量基及其运算法则



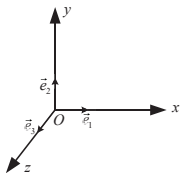
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坐标系矢量基运算法则

$$\vec{e} \times \vec{e}^T = \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix} \times \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix} = \begin{bmatrix} \vec{e}_1 \times \vec{e}_1 & \vec{e}_1 \times \vec{e}_2 & \vec{e}_1 \times \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_1 & \vec{e}_2 \times \vec{e}_2 & \vec{e}_2 \times \vec{e}_3 \\ \vec{e}_3 \times \vec{e}_1 & \vec{e}_3 \times \vec{e}_2 & \vec{e}_3 \times \vec{e}_3 \end{bmatrix}$$

右手笛卡尔直角坐标系矢量基及其运算法则



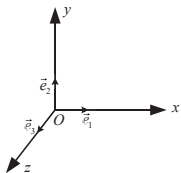
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坐标系矢量基运算法则

$$\vec{e} \times \vec{e}^T = \begin{bmatrix} \vec{e}_1 \times \vec{e}_1 & \vec{e}_1 \times \vec{e}_2 & \vec{e}_1 \times \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_1 & \vec{e}_2 \times \vec{e}_2 & \vec{e}_2 \times \vec{e}_3 \\ \vec{e}_3 \times \vec{e}_1 & \vec{e}_3 \times \vec{e}_2 & \vec{e}_3 \times \vec{e}_3 \end{bmatrix} = \begin{bmatrix} \vec{0} & \vec{e}_3 & -\vec{e}_2 \\ -\vec{e}_3 & \vec{0} & \vec{e}_1 \\ \vec{e}_2 & -\vec{e}_1 & \vec{0} \end{bmatrix}$$

右手笛卡尔直角坐标系矢量基及其运算法则



右手笛卡尔直角坐标系矢量基

$$\vec{e} \triangleq \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}, \quad \vec{e}^T = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix}.$$

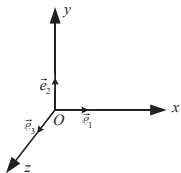
对应任意三维列阵 $\mathbf{u} = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix}^T$, 引入如下定义式

$$\tilde{\mathbf{u}} \triangleq \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}, \quad \tilde{\mathbf{u}}(m, n) \triangleq -\varepsilon_{mnl} u_l,$$

满足

$$\tilde{\mathbf{u}}^T = -\tilde{\mathbf{u}}, \quad \tilde{\mathbf{u}}\mathbf{v} = -\tilde{\mathbf{v}}\mathbf{u}, \quad \mathbf{v}^T \tilde{\mathbf{u}} = -\mathbf{u}^T \tilde{\mathbf{v}}.$$

右手笛卡尔直角坐标系矢量基及其运算法则



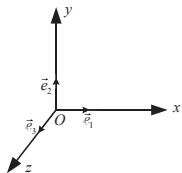
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坐标系矢量基运算法则

$$\vec{e} \times \vec{e}^T = \begin{bmatrix} \vec{e}_1 \times \vec{e}_1 & \vec{e}_1 \times \vec{e}_2 & \vec{e}_1 \times \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_1 & \vec{e}_2 \times \vec{e}_2 & \vec{e}_2 \times \vec{e}_3 \\ \vec{e}_3 \times \vec{e}_1 & \vec{e}_3 \times \vec{e}_2 & \vec{e}_3 \times \vec{e}_3 \end{bmatrix} = \begin{bmatrix} \vec{0} & \vec{e}_3 & -\vec{e}_2 \\ -\vec{e}_3 & \vec{0} & \vec{e}_1 \\ \vec{e}_2 & -\vec{e}_1 & \vec{0} \end{bmatrix} = -\tilde{\vec{e}}$$

右手笛卡尔直角坐标系矢量基及其运算法则



右手笛卡尔直角坐标系矢量基

$$\vec{e} \triangleq \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}, \quad \vec{e}^T = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix}.$$

坐标系矢量基运算法则

(汇总)

$$\vec{e}^T \cdot \vec{e} = 3$$

$$\vec{e} \cdot \vec{e}^T = I_3$$

$$\vec{e}^T \times \vec{e} = \vec{0}$$

$$\vec{e} \times \vec{e}^T = -\tilde{\vec{e}}$$

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矢量的坐标表示

矢量的坐标分量表示形式

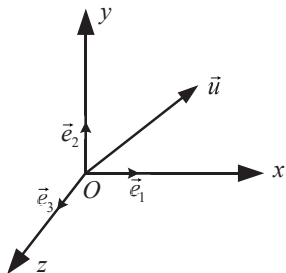
$$\vec{u} = u_1 \vec{e}_1 + u_2 \vec{e}_2 + u_3 \vec{e}_3$$

可记为矩阵乘积 (数乘) 的形式, 即

$$\vec{u} = \begin{bmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} =: \vec{e}^T \mathbf{u} = \mathbf{u}^T \vec{e},$$

式中

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$



称为矢量 $\vec{u}$ 的坐标列阵或投影列阵.

矢量运算的坐标表示

两矢量的点积（内积）：

$$\vec{u} \cdot \vec{v} = \mathbf{u}^T \vec{e} \cdot \vec{e}^T \mathbf{v} = \mathbf{u}^T \mathbf{v}.$$

两矢量的叉积：

$$\vec{u} \times \vec{v} = \mathbf{u}^T \vec{e} \times \vec{e}^T \mathbf{v} = -\mathbf{u}^T \tilde{\mathbf{e}} \mathbf{v} = \vec{e}^T \tilde{\mathbf{u}} \mathbf{v}. \quad (\tilde{\mathbf{u}} =: [\mathbf{u}]_{\times})$$

三矢量的混合积：

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \mathbf{u}^T \vec{e} \cdot (\vec{e}^T \tilde{\mathbf{v}} \mathbf{w}) = \mathbf{u}^T \vec{e} \cdot \vec{e}^T \tilde{\mathbf{v}} \mathbf{w} = \mathbf{u}^T \tilde{\mathbf{v}} \mathbf{w}.$$

三矢量的叉积：

$$\vec{u} \times (\vec{v} \times \vec{w}) = (\vec{e}^T \mathbf{u}) \times (\vec{e}^T \tilde{\mathbf{v}} \mathbf{w}) = \vec{e}^T \tilde{\mathbf{u}} \tilde{\mathbf{v}} \mathbf{w}.$$

叉乘矩阵的性质: $\tilde{\mathbf{u}}^T = -\tilde{\mathbf{u}}, \quad \tilde{\mathbf{u}}\mathbf{v} = -\tilde{\mathbf{v}}\mathbf{u}, \quad \mathbf{v}^T\tilde{\mathbf{u}} = -\mathbf{u}^T\tilde{\mathbf{v}}.$

$$\begin{aligned}
 \tilde{\mathbf{u}}\tilde{\mathbf{v}} &= \begin{bmatrix} 0 & u_3 & -u_2 \\ -u_3 & 0 & u_1 \\ u_2 & -u_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & v_3 & -v_2 \\ -v_3 & 0 & v_1 \\ v_2 & -v_1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} -v_2u_2 - v_3u_3 & v_1u_2 & v_1u_3 \\ v_2u_1 & -v_3u_3 - v_1u_1 & v_2u_3 \\ v_3u_1 & v_3u_2 & -v_1u_1 - v_2u_2 \end{bmatrix} \\
 &= \begin{bmatrix} v_1u_1 & v_1u_2 & v_1u_3 \\ v_2u_1 & v_2u_2 & v_2u_3 \\ v_3u_1 & v_3u_2 & v_3u_3 \end{bmatrix} + (v_1u_1 + v_2u_2 + v_3u_3)\mathbf{I}_3 \\
 &= \mathbf{v}\mathbf{u}^T - (\mathbf{v}^T\mathbf{u})\mathbf{I}_3.
 \end{aligned}$$

三矢量的叉积:

$$\begin{aligned}\vec{u} \times (\vec{v} \times \vec{w}) &= \vec{e}^T \tilde{u} \tilde{v} \tilde{w} \\&= \vec{e}^T \left[\vec{v} \vec{u}^T - (\vec{v}^T \vec{u}) \mathbf{I}_3 \right] \vec{w} \\&= \vec{e}^T \left[\vec{v} \vec{u}^T \vec{w} - (\vec{v}^T \vec{u}) \vec{w} \right] \\&= \vec{e}^T \left[\vec{v} (\vec{u}^T \vec{w}) - \vec{w} (\vec{v}^T \vec{u}) \right] \\&= \vec{e}^T \vec{v} (\vec{u}^T \vec{w}) - \vec{e}^T \vec{w} (\vec{v}^T \vec{u}) \\&= \vec{v} (\vec{u}^T \vec{w}) - \vec{w} (\vec{v}^T \vec{u}) \\&= (\vec{u}^T \vec{w}) \vec{v} - (\vec{v}^T \vec{u}) \vec{w} \\&= (\vec{u} \cdot \vec{w}) \vec{v} - (\vec{u} \cdot \vec{v}) \vec{w}.\end{aligned}$$

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不同坐标系矢量基之间的关系与方向余弦矩阵

两坐标系坐标轴上的单位矢量之间的关系

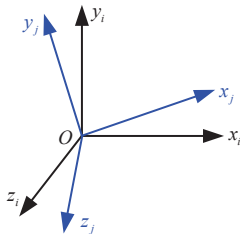
$$\begin{cases} (\vec{e}_1)_i = (a_{11})_{ij}(\vec{e}_1)_j + (a_{12})_{ij}(\vec{e}_2)_j + (a_{13})_{ij}(\vec{e}_3)_j \\ (\vec{e}_2)_i = (a_{21})_{ij}(\vec{e}_1)_j + (a_{22})_{ij}(\vec{e}_2)_j + (a_{23})_{ij}(\vec{e}_3)_j \\ (\vec{e}_3)_i = (a_{31})_{ij}(\vec{e}_1)_j + (a_{32})_{ij}(\vec{e}_2)_j + (a_{33})_{ij}(\vec{e}_3)_j \end{cases}$$

矩阵形式

$$\begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}_i = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{ij} \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}_j$$

式中

$$(a_{mn})_{ij} = (\vec{e}_m)_i \cdot (\vec{e}_n)_j = \cos \langle (\vec{e}_m)_i, (\vec{e}_n)_j \rangle$$



$$\begin{cases} \vec{e}_i = A_{ij} \vec{e}_j \\ A_{ij} = \vec{e}_i \cdot \vec{e}_j^T \end{cases}$$

方向余弦矩阵的性质

(1) 两个方向相同的坐标系之间的方向余弦矩阵为单位矩阵.

证明：当坐标系 i 和坐标系 j 方向相同时，有

$$\vec{e}_j = \vec{e}_i,$$

则两者之间的方向余弦矩阵满足

$$\begin{aligned} A_{ij} &= \vec{e}_i \cdot \vec{e}_j^T \\ &= \vec{e}_i \cdot \vec{e}_i^T \\ &= I_3. \end{aligned}$$

方向余弦矩阵的性质

(2) 方向余弦矩阵中的 9 个数满足 6 个独立的约束方程

证明：用坐标系 j 的矢量基表示的坐标系 i 的矢量基

$$\begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}_i = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{ij} \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}_j$$

满足 (I) i 系坐标轴上的单位矢量模为 1:

$$\begin{cases} |(\vec{e}_1)|_i = \sqrt{a_{11}^2 + a_{12}^2 + a_{13}^2} = 1 \\ |(\vec{e}_2)|_i = \sqrt{a_{21}^2 + a_{22}^2 + a_{23}^2} = 1 \\ |(\vec{e}_3)|_i = \sqrt{a_{31}^2 + a_{32}^2 + a_{33}^2} = 1 \end{cases}$$

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满足 (II) i 系坐标轴上的单位矢量两两正交：

$$\begin{cases} |(\vec{e}_1)|_i \cdot |(\vec{e}_2)|_i = a_{11}a_{21} + a_{12}a_{22} + a_{13}a_{23} = 0 \\ |(\vec{e}_2)|_i \cdot |(\vec{e}_3)|_i = a_{21}a_{31} + a_{22}a_{32} + a_{23}a_{33} = 0 \\ |(\vec{e}_3)|_i \cdot |(\vec{e}_1)|_i = a_{31}a_{11} + a_{32}a_{12} + a_{33}a_{13} = 0 \end{cases}$$

方向余弦矩阵的性质

(3) 方向余弦矩阵等于其余子矩阵，即方向余弦矩阵中的每个数都等于其对应的代数余子式，即

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{ij} = \begin{bmatrix} - \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{12} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ - \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}.$$

证明:

$$\begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}_i = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{ij} \begin{bmatrix} \vec{e}_1 \\ \vec{e}_2 \\ \vec{e}_3 \end{bmatrix}_j$$

$$(\vec{e}_1)_i = (\vec{e}_2)_i \times (\vec{e}_3)_i$$

$$\begin{aligned} &= \begin{vmatrix} (\vec{e}_1)_j & (\vec{e}_2)_j & (\vec{e}_3)_j \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} (\vec{e}_1)_j - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} (\vec{e}_2)_j + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} (\vec{e}_3)_j \\ &= a_{11}(\vec{e}_1)_j + a_{12}(\vec{e}_2)_j + a_{13}(\vec{e}_3)_j \end{aligned}$$

方向余弦矩阵的性质

(4) 方向余弦矩阵行列式的值为正 1.

证明:

$$\begin{aligned} A_{ij} &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}_{ij} \\ &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}^2 + a_{12}^2 + a_{13}^2 \\ &= 1. \end{aligned}$$

方向余弦矩阵的性质

(5) 方向余弦矩阵为正交矩阵，逆矩阵等于其转置.

证明:

$$\begin{aligned} I_3 &= \vec{e}_i \cdot \vec{e}_i^T \\ &= (A_{ij} \vec{e}_j) \cdot (A_{ij} \vec{e}_j)^T \\ &= A_{ij} \vec{e}_j \cdot \vec{e}_j^T A_{ij}^T \\ &= A_{ij} I_3 A_{ij}^T \\ &= A_{ij} A_{ij}^T, \end{aligned}$$

由此可得

$$A_{ij}^{-1} = A_{ij}^T.$$

方向余弦矩阵的性质

(6) 同一矢量 $\vec{r}$ 在 i 中的投影列阵 $\mathbf{r}_i$ 和它在 j 系中的投影列阵 $\mathbf{r}_j$ 之间满足坐标变换式 $\mathbf{r}_i = \mathbf{A}_{ij}\mathbf{r}_j$.

证明:

$$\begin{aligned}\vec{r} &= \vec{e}_i^T \mathbf{r}_i = \vec{e}_j^T \mathbf{r}_j \\ &= (\mathbf{A}_{ij} \vec{e}_j)^T \mathbf{r}_i \\ &= \vec{e}_j^T \mathbf{A}_{ij}^T \mathbf{r}_i,\end{aligned}$$

由此可得

$$\mathbf{r}_j = \mathbf{A}_{ij}^T \mathbf{r}_i \quad \Rightarrow \quad \mathbf{r}_i = \mathbf{A}_{ij} \mathbf{r}_j.$$

方向余弦矩阵的性质

(7) 同一矢量 $\vec{r}$ 在 i 中的投影列阵 $\mathbf{r}_i$ 对应的叉乘矩阵 $\tilde{\mathbf{r}}_i$ 和它在 j 系中的投影列阵 $\mathbf{r}_j$ 对应的叉乘矩阵 $\tilde{\mathbf{r}}_j$ 之间满足如下坐标变换式

$$\tilde{\mathbf{r}}_i = \mathbf{A}_{ij} \tilde{\mathbf{r}}_j \mathbf{A}_{ij}^T,$$

由此可得

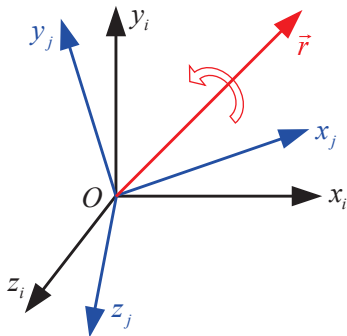
$$\mathbf{A}_{ij}^T \tilde{\mathbf{r}}_i = \tilde{\mathbf{r}}_j \mathbf{A}_{ij}^T$$

和

$$\tilde{\mathbf{r}}_i \mathbf{A}_{ij} = \mathbf{A}_{ij} \tilde{\mathbf{r}}_j.$$

方向余弦矩阵的性质

(8) 任意两个坐标系 i 和 j 之间的方向余弦矩阵 $\mathbf{A}_{ij}$ 都有一个实特征值 1, 即存在一个实数形式的特征向量 $\mathbf{r}$, 该特征向量满足 $\mathbf{r} = \mathbf{A}_{ij}\mathbf{r}$. 该性质表明两坐标间的任意相对转动都可以通过一次定轴 (简单) 转动来完成.

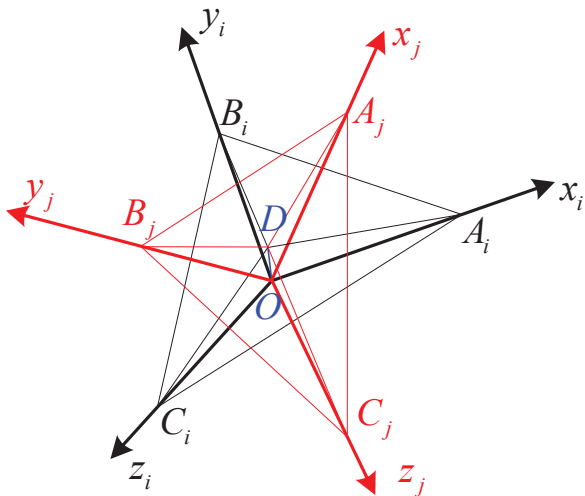


证明：在不引起歧义的前提下将方向余弦矩阵 $\mathbf{A}_{ij}$ 简记为 $\mathbf{A}$. 其特征多项式为

$$\begin{aligned} f_{\mathbf{A}}(\lambda) &= \det(\mathbf{A} - \lambda \mathbf{I}_3) = |\mathbf{A} - \lambda \mathbf{I}_3| \\ &= \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix} \\ &= -\lambda^3 + \operatorname{tr}(\mathbf{A})\lambda^2 - \sum_{k=1}^3 M_{kk}\lambda + \det(\mathbf{A}) \\ &= -\lambda^3 + \operatorname{tr}(\mathbf{A})\lambda^2 - \operatorname{tr}(\mathbf{A})\lambda + 1. \end{aligned}$$

将 $\lambda = 1$ 代入上式可得

$$f_{\mathbf{A}}(1) = -1 + \operatorname{tr}(\mathbf{A}) - \operatorname{tr}(\mathbf{A}) + 1 = 0.$$

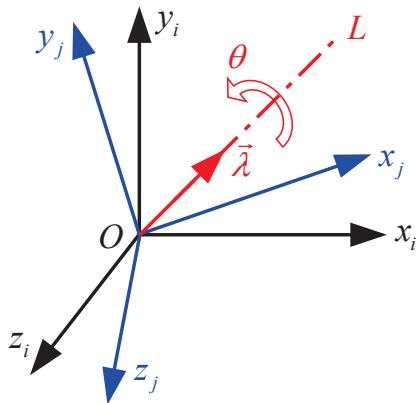


Tip: $\angle A_iDA_j = \angle B_iDB_j = \angle C_iDC_j$

刚体的空间运动 -I

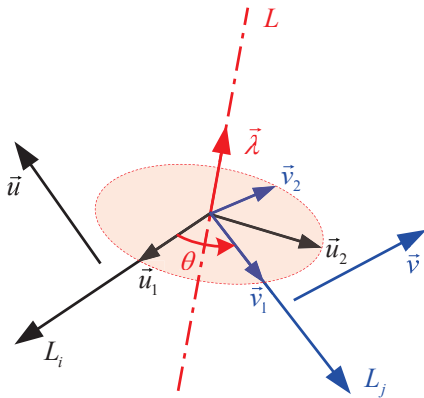
- ① 坐标系、坐标轴上的单位矢量与矢量基
- ② 矢量及矢量运算的坐标表示
- ③ 不同坐标系矢量基之间的关系与方向余弦矩阵
- ④ 刚体的定点简单转动
- ⑤ 定点简单转动与方向余弦矩阵的关系
- ⑥ 欧拉参数

刚体的定点简单转动



两坐标系的相对方位与转轴单位矢量 $\vec{\lambda}$ 和转角 θ 的关系？

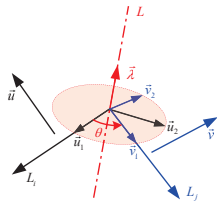
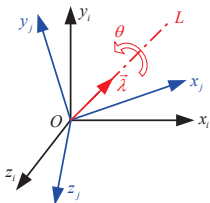
刚体的定点简单转动



$$\vec{v} = \cos \theta \vec{u} + \sin \theta \vec{\lambda} \times \vec{u} + (1 - \cos \theta) \vec{\lambda} \vec{\lambda} \cdot \vec{u}$$

刚体的空间运动 -I

- ① 坐标系、坐标轴上的单位矢量与矢量基
- ② 矢量及矢量运算的坐标表示
- ③ 不同坐标系矢量基之间的关系与方向余弦矩阵
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- ⑤ 定点简单转动与方向余弦矩阵的关系
- ⑥ 欧拉参数



$$\vec{v} = \cos \theta \vec{u} + \sin \theta \vec{\lambda} \times \vec{u} + (1 - \cos \theta) \vec{\lambda} \vec{\lambda} \cdot \vec{u}$$

上述等式两端分别向 j 系和 i 系投影可得

$$\begin{aligned} \vec{e}_j^T \vec{v}_j &= \vec{e}_i^T \vec{u}_i \cos \theta + \vec{e}_i^T \tilde{\lambda}_i \vec{u}_i \sin \theta + \vec{e}_i^T \lambda_i \lambda_i^T \vec{u}_i (1 - \cos \theta) \\ &= \vec{e}_i^T \left[\mathbf{I}_3 \cos \theta + \tilde{\lambda}_i \sin \theta + \lambda_i \lambda_i^T (1 - \cos \theta) \right] \vec{u}_i. \end{aligned}$$

上页结论:

$$\vec{e}_j^T \mathbf{v}_j = \vec{e}_i^T \left[I_3 \cos \theta + \tilde{\boldsymbol{\lambda}}_i \sin \theta + \boldsymbol{\lambda}_i \boldsymbol{\lambda}_i^T (1 - \cos \theta) \right] \mathbf{u}_i. \quad (\text{I})$$

注意到

$$\vec{e}_j^T = (\mathbf{A}_{ji} \vec{e}_i)^T = \vec{e}_i^T \mathbf{A}_{ji}^T = \vec{e}_i^T \mathbf{A}_{ij},$$

$$\mathbf{v}_j = \mathbf{u}_i,$$

$$\boldsymbol{\lambda}_j = \boldsymbol{\lambda}_i =: \boldsymbol{\lambda},$$

将其代入 (I) 式可得

$$\vec{e}_i^T \mathbf{A}_{ij} \mathbf{u}_i = \vec{e}_i^T \left[I_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \right] \mathbf{u}_i.$$

移项后整理可得

$$\vec{e}_i^T \left\{ \mathbf{A}_{ij} - \left[I_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \right] \right\} \mathbf{u}_i = \vec{0}.$$

由

$$\vec{e}_i^T \left\{ \mathbf{A}_{ij} - \left[\mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \right] \right\} \mathbf{u}_i = \vec{0}$$

可得

$$\left\{ \mathbf{A}_{ij} - \left[\mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \right] \right\} \mathbf{u}_i = \mathbf{0}.$$

进一步, 由 $\mathbf{u}_i$ 的任意性可得

$$\mathbf{A}_{ij} - \left[\mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \right] = \mathbf{O}_{3 \times 3}.$$

由此可得方向余弦矩阵与转轴转角的关系式, 即

$$\begin{aligned} \mathbf{A}_{ij} &= \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \\ &= \begin{bmatrix} \cos \theta + \lambda_1 \lambda_1 (1 - \cos \theta) & -\lambda_3 \sin \theta + \lambda_1 \lambda_2 (1 - \cos \theta) & \lambda_2 \sin \theta + \lambda_1 \lambda_3 (1 - \cos \theta) \\ \lambda_3 \sin \theta + \lambda_2 \lambda_1 (1 - \cos \theta) & \cos \theta + \lambda_2 \lambda_2 (1 - \cos \theta) & -\lambda_1 \sin \theta + \lambda_2 \lambda_3 (1 - \cos \theta) \\ -\lambda_2 \sin \theta + \lambda_3 \lambda_1 (1 - \cos \theta) & \lambda_1 \sin \theta + \lambda_3 \lambda_2 (1 - \cos \theta) & \cos \theta + \lambda_3 \lambda_3 (1 - \cos \theta) \end{bmatrix}. \end{aligned}$$

对于微小转动 ($|\theta| \ll 1$), 有

$$\mathbf{A}_{ij} \approx \mathbf{I}_3 + \tilde{\boldsymbol{\lambda}} \theta = \mathbf{I}_3 + \widetilde{\boldsymbol{\lambda} \theta} =: \mathbf{I}_3 + \tilde{\boldsymbol{\theta}}_{ij}.$$

方向余弦矩阵与转轴转角的关系式:

$$\begin{aligned} \mathbf{A}_{ij} &= \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \\ &= \begin{bmatrix} \cos \theta + \lambda_1 \lambda_1 (1 - \cos \theta) & -\lambda_3 \sin \theta + \lambda_1 \lambda_2 (1 - \cos \theta) & \lambda_2 \sin \theta + \lambda_1 \lambda_3 (1 - \cos \theta) \\ \lambda_3 \sin \theta + \lambda_2 \lambda_1 (1 - \cos \theta) & \cos \theta + \lambda_2 \lambda_2 (1 - \cos \theta) & -\lambda_1 \sin \theta + \lambda_2 \lambda_3 (1 - \cos \theta) \\ -\lambda_2 \sin \theta + \lambda_3 \lambda_1 (1 - \cos \theta) & \lambda_1 \sin \theta + \lambda_3 \lambda_2 (1 - \cos \theta) & \cos \theta + \lambda_3 \lambda_3 (1 - \cos \theta) \end{bmatrix}. \end{aligned}$$

(I) 当 $\boldsymbol{\lambda} = [\lambda_1 \quad \lambda_2 \quad \lambda_3]^T = [1 \quad 0 \quad 0]^T$ 时,

$$\mathbf{A}_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} =: \mathbf{A}_x(\theta).$$

方向余弦矩阵与转轴转角的关系式:

$$\begin{aligned} \mathbf{A}_{ij} &= \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \\ &= \begin{bmatrix} \cos \theta + \lambda_1 \lambda_1 (1 - \cos \theta) & -\lambda_3 \sin \theta + \lambda_1 \lambda_2 (1 - \cos \theta) & \lambda_2 \sin \theta + \lambda_1 \lambda_3 (1 - \cos \theta) \\ \lambda_3 \sin \theta + \lambda_2 \lambda_1 (1 - \cos \theta) & \cos \theta + \lambda_2 \lambda_2 (1 - \cos \theta) & -\lambda_1 \sin \theta + \lambda_2 \lambda_3 (1 - \cos \theta) \\ -\lambda_2 \sin \theta + \lambda_3 \lambda_1 (1 - \cos \theta) & \lambda_1 \sin \theta + \lambda_3 \lambda_2 (1 - \cos \theta) & \cos \theta + \lambda_3 \lambda_3 (1 - \cos \theta) \end{bmatrix}. \end{aligned}$$

(II) 当 $\boldsymbol{\lambda} = [\lambda_1 \quad \lambda_2 \quad \lambda_3]^T = [0 \quad 1 \quad 0]^T$ 时,

$$\mathbf{A}_{ij} = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} =: \mathbf{A}_y(\theta).$$

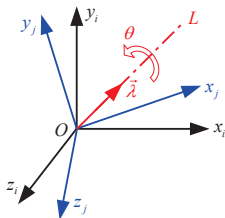
方向余弦矩阵与转轴转角的关系式:

$$\begin{aligned}\mathbf{A}_{ij} &= \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \\ &= \begin{bmatrix} \cos \theta + \lambda_1 \lambda_1 (1 - \cos \theta) & -\lambda_3 \sin \theta + \lambda_1 \lambda_2 (1 - \cos \theta) & \lambda_2 \sin \theta + \lambda_1 \lambda_3 (1 - \cos \theta) \\ \lambda_3 \sin \theta + \lambda_2 \lambda_1 (1 - \cos \theta) & \cos \theta + \lambda_2 \lambda_2 (1 - \cos \theta) & -\lambda_1 \sin \theta + \lambda_2 \lambda_3 (1 - \cos \theta) \\ -\lambda_2 \sin \theta + \lambda_3 \lambda_1 (1 - \cos \theta) & \lambda_1 \sin \theta + \lambda_3 \lambda_2 (1 - \cos \theta) & \cos \theta + \lambda_3 \lambda_3 (1 - \cos \theta) \end{bmatrix}.\end{aligned}$$

(III) 当 $\boldsymbol{\lambda} = [\lambda_1 \quad \lambda_2 \quad \lambda_3]^T = [0 \quad 0 \quad 1]^T$ 时,

$$\mathbf{A}_{ij} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} =: \mathbf{A}_z(\theta).$$

定点简单转动与方向余弦矩阵的关系



(I) 绕 x 轴转动 ($\lambda = [1 \ 0 \ 0]^T$):

$$A_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}.$$

(II) 绕 y 轴转动 ($\lambda = [0 \ 1 \ 0]^T$):

$$A_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}.$$

$A_{ij} = I_3 \cos \theta + \tilde{\lambda} \sin \theta + \lambda \lambda^T (1 - \cos \theta)$ (III) 绕 z 轴转动 ($\lambda = [0 \ 0 \ 1]^T$):

$$A_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

方向余弦矩阵与转轴和转角的关系式:

$$\begin{aligned} \mathbf{A}_{ij} &= \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta) \\ &= \begin{bmatrix} \cos \theta + \lambda_1 \lambda_1 (1 - \cos \theta) & -\lambda_3 \sin \theta + \lambda_1 \lambda_2 (1 - \cos \theta) & \lambda_2 \sin \theta + \lambda_1 \lambda_3 (1 - \cos \theta) \\ \lambda_3 \sin \theta + \lambda_2 \lambda_1 (1 - \cos \theta) & \cos \theta + \lambda_2 \lambda_2 (1 - \cos \theta) & -\lambda_1 \sin \theta + \lambda_2 \lambda_3 (1 - \cos \theta) \\ -\lambda_2 \sin \theta + \lambda_3 \lambda_1 (1 - \cos \theta) & \lambda_1 \sin \theta + \lambda_3 \lambda_2 (1 - \cos \theta) & \cos \theta + \lambda_3 \lambda_3 (1 - \cos \theta) \end{bmatrix} \\ &=: \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \end{aligned}$$

由上式可知:

$$\begin{aligned} \text{tr} \mathbf{A} &= a_{11} + a_{22} + a_{33} \\ &= 3 \cos \theta + (\lambda_1^2 + \lambda_2^2 + \lambda_3^2)(1 - \cos \theta) \\ &= 2 \cos \theta + 1 \end{aligned} \quad \left\{ \begin{array}{l} a_{32} - a_{23} = 2\lambda_1 \sin \theta \\ a_{13} - a_{31} = 2\lambda_2 \sin \theta \\ a_{21} - a_{12} = 2\lambda_3 \sin \theta \end{array} \right.$$

方向余弦矩阵与转轴和转角的关系式:

$$\mathbf{A}_{ij} = \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta), \quad \begin{cases} \cos \theta &= (\text{tr} \mathbf{A} - 1)/2, \\ \lambda_1 \sin \theta &= (a_{32} - a_{23})/2, \\ \lambda_2 \sin \theta &= (a_{13} - a_{31})/2, \\ \lambda_3 \sin \theta &= (a_{21} - a_{12})/2. \end{cases}$$

若约定 $\theta \in [0, \pi]$, 则 (1) 当

$$|(\text{tr} \mathbf{A} - 1)/2 - 1| < \varepsilon$$

时

$$\begin{cases} \cos \theta \approx 1, \\ \sin \theta \approx 0, \end{cases} \Rightarrow \begin{cases} \theta \approx 0, \\ \mathbf{A} \approx \mathbf{I}_3, \end{cases}$$

转轴 $\boldsymbol{\lambda}$ 可为任意单位矢量列阵.

方向余弦矩阵与转轴和转角的关系式:

$$\mathbf{A}_{ij} = \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta), \quad \left\{ \begin{array}{l} \cos \theta = (\text{tr} \mathbf{A} - 1)/2, \\ \lambda_1 \sin \theta = (a_{32} - a_{23})/2, \\ \lambda_2 \sin \theta = (a_{13} - a_{31})/2, \\ \lambda_3 \sin \theta = (a_{21} - a_{12})/2. \end{array} \right.$$

若约定 $\theta \in [0 \ \pi]$, 则 (2) 当

$$|(\text{tr} \mathbf{A} - 1)/2 + 1| < \varepsilon$$

时

$$\left\{ \begin{array}{l} \cos \theta \approx -1, \\ \sin \theta \approx 0, \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \theta \approx \pi, \\ (\mathbf{A} + \mathbf{I})/2 \approx \boldsymbol{\lambda} \boldsymbol{\lambda}^T, \end{array} \right.$$

依此确定 $\boldsymbol{\lambda}$ 的按模最大分量后即可求出完整的 $\boldsymbol{\lambda}$.

方向余弦矩阵与转轴和转角的关系式:

$$\mathbf{A}_{ij} = \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta), \quad \begin{cases} \cos \theta &= (\operatorname{tr} \mathbf{A} - 1)/2, \\ \lambda_1 \sin \theta &= (a_{32} - a_{23})/2, \\ \lambda_2 \sin \theta &= (a_{13} - a_{31})/2, \\ \lambda_3 \sin \theta &= (a_{21} - a_{12})/2. \end{cases}$$

若约定 $\theta \in [0, \pi]$, 则 (3) 当

$$|(\operatorname{tr} \mathbf{A} - 1)/2 - 1| > \varepsilon \quad \text{and} \quad |(\operatorname{tr} \mathbf{A} - 1)/2 + 1| > \varepsilon$$

时

$$\begin{cases} \theta &= \arccos((\operatorname{tr} \mathbf{A} - 1)/2), \\ \lambda_1 &= (a_{32} - a_{23})/(2 \sin \theta), \\ \lambda_2 &= (a_{13} - a_{31})/(2 \sin \theta), \\ \lambda_3 &= (a_{21} - a_{12})/(2 \sin \theta). \end{cases}$$

刚体的空间运动 -I

- ① 坐标系、坐标轴上的单位矢量与矢量基
- ② 矢量及矢量运算的坐标表示
- ③ 不同坐标系矢量基之间的关系与方向余弦矩阵
- ④ 刚体的定点简单转动
- ⑤ 定点简单转动与方向余弦矩阵的关系
- ⑥ 欧拉参数

欧拉参数

方向余弦矩阵与转轴和转角的关系式:

$$\mathbf{A}_{ij} = \mathbf{I}_3 \cos \theta + \tilde{\boldsymbol{\lambda}} \sin \theta + \boldsymbol{\lambda} \boldsymbol{\lambda}^T (1 - \cos \theta).$$

三角函数的倍角公式:

$$\cos \theta = 2 \cos^2 \frac{\theta}{2} - 1, \quad \sin \theta = 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2}, \quad (1 - \cos \theta) = 2 \sin^2 \frac{\theta}{2}.$$

定义四个欧拉参数

$$\begin{bmatrix} \varepsilon_1 & \varepsilon_2 & \varepsilon_3 \end{bmatrix}^T \triangleq \boldsymbol{\lambda} \sin \frac{\theta}{2} =: \bar{\boldsymbol{\varepsilon}}, \quad \varepsilon_4 \triangleq \cos \frac{\theta}{2},$$

则有如下多项式矩阵形式的方向余弦矩阵

$$\mathbf{A}_{ij} = \mathbf{I}_3 (2\varepsilon_4^2 - 1) + 2\tilde{\boldsymbol{\varepsilon}}\varepsilon_4 + 2\bar{\boldsymbol{\varepsilon}}\bar{\boldsymbol{\varepsilon}}^T.$$

欧拉参数

四个欧拉参数

$$\begin{bmatrix} \varepsilon_1 & \varepsilon_2 & \varepsilon_3 \end{bmatrix}^T \triangleq \boldsymbol{\lambda} \sin \frac{\theta}{2} =: \bar{\boldsymbol{\varepsilon}}, \quad \varepsilon_4 \triangleq \cos \frac{\theta}{2},$$

满足如下约束方程：

$$\bar{\boldsymbol{\varepsilon}}^2 + \varepsilon_4^2 = \boldsymbol{\lambda}^2 \sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} = \sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} = 1.$$

则用欧拉参数表示的方向余弦矩阵形式不唯一：

$$\begin{cases} \mathbf{A}_{ij} = \mathbf{I}_3(2\varepsilon_4^2 - 1) + 2\tilde{\boldsymbol{\varepsilon}}\varepsilon_4 + 2\bar{\boldsymbol{\varepsilon}}\bar{\boldsymbol{\varepsilon}}^T, \\ \mathbf{A}_{ij} = \mathbf{I}_3(1 - 2\bar{\boldsymbol{\varepsilon}}^2) + 2\tilde{\boldsymbol{\varepsilon}}\varepsilon_4 + 2\bar{\boldsymbol{\varepsilon}}\bar{\boldsymbol{\varepsilon}}^T. \end{cases}$$

欧拉参数

用欧拉参数

$$\begin{bmatrix} \varepsilon_1 & \varepsilon_2 & \varepsilon_3 \end{bmatrix}^T \triangleq \boldsymbol{\lambda} \sin \frac{\theta}{2} =: \bar{\boldsymbol{\varepsilon}}, \quad \varepsilon_4 \triangleq \cos \frac{\theta}{2}$$

表示的方向余弦矩阵

$$\begin{aligned} \mathbf{A}_{ij} &= \mathbf{I}_3(1 - 2\bar{\boldsymbol{\varepsilon}}^2) + 2\tilde{\boldsymbol{\varepsilon}}\varepsilon_4 + 2\bar{\boldsymbol{\varepsilon}}\bar{\boldsymbol{\varepsilon}}^T \\ &= \begin{bmatrix} 1 - 2\varepsilon_2^2 - 2\varepsilon_3^2 & 2(\varepsilon_1\varepsilon_2 - \varepsilon_3\varepsilon_4) & 2(\varepsilon_1\varepsilon_3 + \varepsilon_2\varepsilon_4) \\ 2(\varepsilon_2\varepsilon_1 + \varepsilon_3\varepsilon_4) & 1 - 2\varepsilon_3^2 - 2\varepsilon_1^2 & 2(\varepsilon_2\varepsilon_3 - \varepsilon_1\varepsilon_4) \\ 2(\varepsilon_3\varepsilon_1 - \varepsilon_2\varepsilon_4) & 2(\varepsilon_3\varepsilon_2 + \varepsilon_1\varepsilon_4) & 1 - 2\varepsilon_1^2 - 2\varepsilon_2^2 \end{bmatrix}. \end{aligned}$$